

Table 1. *G36-Cyber* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Medium-sized office building	Chicago, IL, USA	Simulated	G36	6	114	5-min	1 day

Table 2. Details of *G36-Cyber* fault scenarios. Fault injection times varied in each case.

Category	Fault abbreviation (CSV file name)	Fault type	Fault intensity	Duration	Method of fault imposition
Data-intrusion attack	Intru_TRstReq_Max	Temperature reset request	15 temperature reset request	12 p.m. to 3 p.m. on one shoulder season day (with the next 2 hours as the post-attack period).	Corrupting the number of transmitted zone temperature reset requests to 15 for five thermal zones (maximum allowed in G36) using the <i>Max temporal</i> model: $\hat{y}(t) = \begin{cases} y(t), & t \notin A \\ y_{max}, & t \in A \end{cases}$ where $\hat{y}$ is the corrupted property value, y is the original value, and A is the threat period
	Intru_ChillerOnOff	Chiller on/off	Chiller cycles on/off every 30 minutes	12 p.m. to 3 p.m. on one cooling season day (with 3 p.m. to 7 p.m. as the post-attack period).	Using <i>square pulse</i> model with a period of one hour on chiller on/off control signal: $\hat{y}(t) = \begin{cases} y(t), & t \notin A \\ f_p(t), & t \in A \text{ and } c \in [y_{min}, y_{max}] \\ y_{min}, & t \in A \text{ and } c < y_{min} \\ y_{max}, & t \in A \text{ and } c > y_{max} \end{cases}$
	Intru_SupFanSpd_Max	Supply air fan speed at max value	speed set to 100%	7 a.m. to 7 p.m. on one cooling season day.	Using the <i>max temporal</i> model.

	Intru_CoolSP_Const	Zone temperature cooling setpoint to constant 22 °C	-	1 p.m. to 3 p.m. on one cooling season day (with the next 2 hours as the post attack period).	Using the <i>constant temporal</i> model, where property value is overwritten to a user-defined constant value c during the threat: $\hat{y}(t) = \begin{cases} y(t), t \notin A \\ c, t \in A \text{ and } c \in [y_{min}, y_{max}] \\ y_{min}, t \in A \text{ and } c < y_{min} \\ y_{max}, t \in A \text{ and } c > y_{max} \end{cases}$
Denial-of-service (DoS) attach	Dos_CHWSP_Max	Blocking chilled water setpoint	The chiller supply chilled water (CHWS) temperature was continuously reset from a minimum value of 5 °C to a maximum value of 10 °C.	12 p.m. to 6 p.m. on one cooling season day (6 p.m. to 7 p.m. as the post-attack period).	Blocking the chiller from receiving its setpoints to use values from the previous time step, by implementing the <i>blocking temporal</i> model: $\hat{y}(t) = \begin{cases} y(t), t \notin A \\ y(t-1), t \in A \end{cases}$
	DoS_CHWSP_Delay	Delaying chilled water setpoint	Fixed 10-minute delay in communication network	12 p.m. to 6 p.m. on one cooling season day.	Transmitted signal is delayed to the receivers by the <i>delaying</i> model: $\hat{y}(t) = \begin{cases} y(t), t \notin A \\ y(t-\Delta t), t \in A \end{cases}$

Additional information

- Additional helpful document(s):
 - Fu, Y. et al. Modeling and evaluation of cyber-attacks on grid-interactive efficient buildings. *Applied Energy* 303, 117639 (2021).
 - Fu, Y., O'Neill, Z. & Adetola, V. A flexible and generic functional mock-up unit based threat injection framework for grid-interactive efficient buildings: A case study in Modelica. *Energy and Buildings* 250, 111263 (2021).
- In the 'Duration' column of Table , the cooling season day corresponds to the 207th day of the year (July 26th), while the shoulder season day corresponds to the 83rd day of the year (March 24th). Correspondingly, this folder contains two baseline datasets: One for the cooling season day (BaselineSystem_Summer.csv) and one for the shoulder season day (BaselineSystem_Shoulder.csv).

- Please note, since this dataset represents a simulated building, the baseline (fault-free) data was intentionally generated with the same timestamp as the fault data. This was achieved by applying identical outdoor conditions (e.g., outdoor air humidity and temperature), occupancy load, time of day, simulation period, and other related parameters when generating the dataset from the simulated model. This approach allows researchers to compare the corresponding baseline and fault/attack datasets without confounding variations from external factors.
- Fig. 1 shows the schematic of the corresponding AHU system. The system is a single-duct multi-zone VAV AHU connected with five VAV terminal boxes that serve five zones (four exterior and one interior zone) on a single floor.

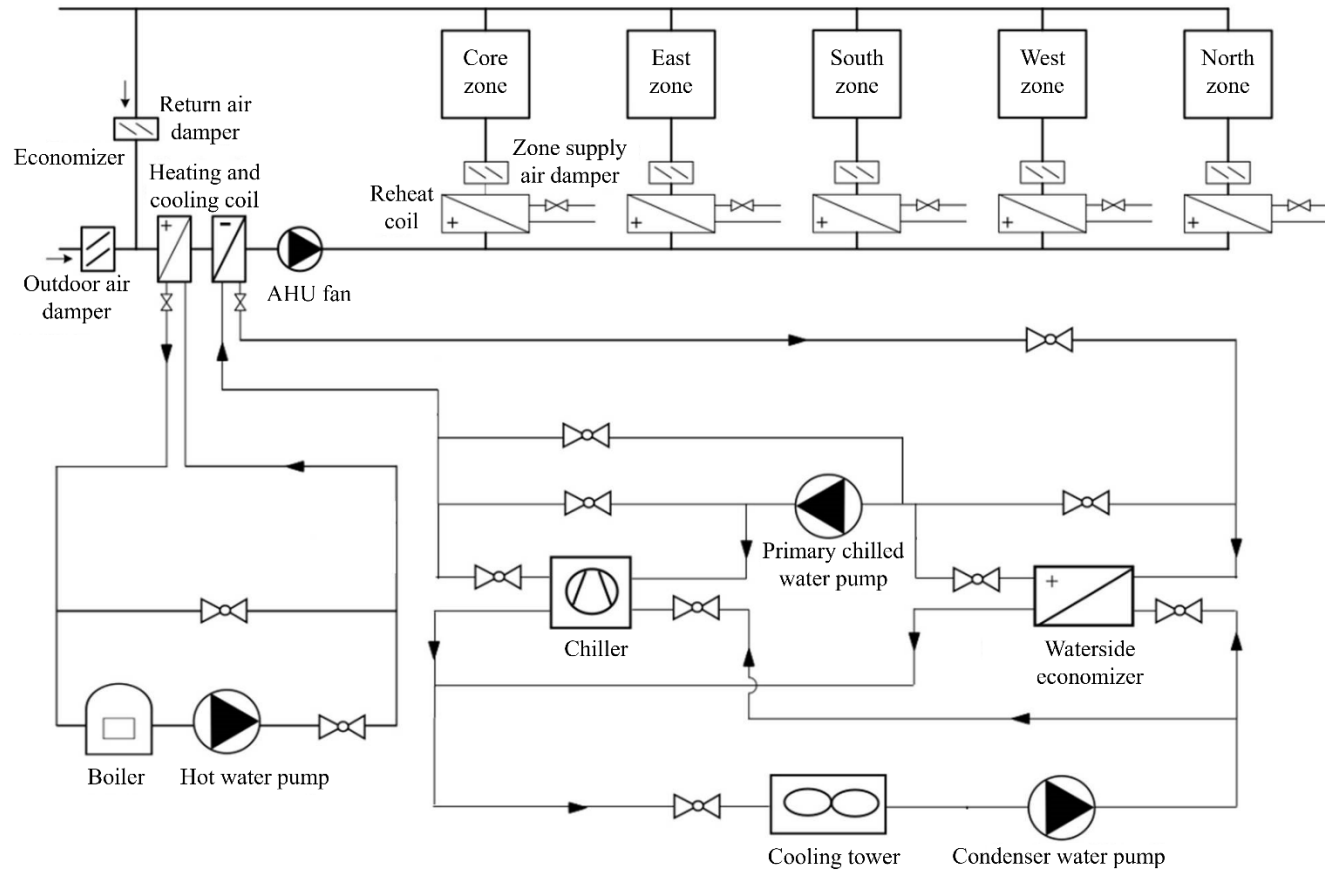


Fig. 1. Schematic of the AHU diagram, used to generate the dataset.